Mass multiplication of *Azolla*

Small plots (50-100 sq.m) or concrete tanks with standing water are inoculated with *Azolla*

↓

pH is adjusted to 8.0 with lime.

↓

Superphosphate is applied as a nutrient and carbofuran is applied to deter insects

↓

After 20 days, *Azolla* biomass can be harvested and used as biofertilizer for rice plantings

Method of application of *Azolla* in rice fields

Azolla is grown on the flooded rice fields prior to planting for 2-3weeks. Then water is drained and *Azolla* is incorporated into the soil followed by rice transplantation within a week's time.

HOTS

Why bio- fertilizer are preferred to chemical fertilizer?

11.3 Biopesticides

Pests are insects that damage crop plants and stored products. They feed on leaves and roots or suck the sap of the plants causing severe crop losses. Chemical pesticides sprayed on plants can be detrimental to ecosystem if the usage is prolonged and pests may develop resistance to the pesticides.

The term **biopesticides** refers to compounds that are used to manage agricultural pests by means of specific biological effects. It refers to products containing biocontrol agents like natural substances such as plants, certain minerals, animals, micro organisms including their genes or metabolites.

They are an important part of Integrated Pest Management (IPM) strategy in controlling the pest.

Advantages

They are less toxic to humans and environment and they do not leave harmful residues.

They affect only the target pest.

They cause long term suppression of pest populations since they persist in the environment.

Microbial biopesticides are of three kinds

1. Bacterial biopesticide
2. Fungal biopesticide
3. Viral biopesticide

11.3.1 Bacterial Biopesticide

Bacteria like *Bacillus thuringiensis, Bacillus papillae* and *Bacillus lentimorbus* have the potential to kill certain insect pests and are entomopathogenic.

Bacillus thuringiensis

It is a gram positive, spore forming, rod shaped soil bacterium. During sporulation, the bacterium produces insecticidal proteins as parasporal crystals. These are called ***delta endotoxin*** also called as ***Cry proteins***. ***Cry proteins*** are specifically

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